Components of a type-theoretic approach to unsupervised learning

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Key Question

How can we frame learning in a way so that we can:

- reason abstractly about learning,
- easily implement theoretical results, and
- draw on existing mathematical knowledge?

A General Model of Learning

Learning can be defined as *the increased performance of some task as experience increases*.[1] This is a model of a learning system:

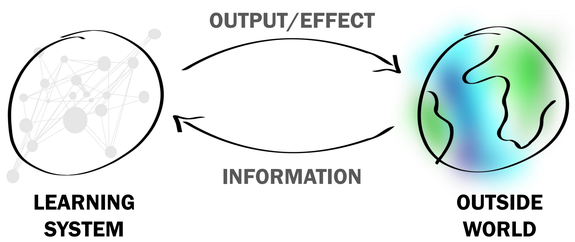

Figure 1. The learning system will likely have a *processor* that can deal with limited amounts of information at a time and a *memory* that can store the information. Performance is likely related to the ability to pull out the immediately relevant bits of information from the outside world.

Perception-Based Model of Learning

For now, we will focus on a learning system that only takes in a stream of information from the outside world; it does not produce an output or an effect. Its goal is to quickly understand a torrent of *primitive* data and synthesize it into useful *abstractions* conveying essentially the same information.

It is the goal in this formulation to draw connections with the constructivist homotopy type theory program, and to allow for the adoption of existing techniques and knowledge from algebraic topology and mathematical physics.

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Toy Model: Object Recognition

SETUP. Consider the following learning system: a computer is connected to a camera, which we think of as a collection of photosensors.

- When light hits a sensor, the color value and sensor index number is transmitted to the computer.
- Assume the computer does not know how the sensors are physically arranged.

FORMULATION. Denote the collection of sensors by O. We can think of reality as a collection of possible states, $\omega \in \Omega$. Then, each state can be thought of as a continuous function,

$$\omega\colon O \to \mathbb{R}.$$

Equivalently, we can write $\omega\colon O \times \mathbb{R}$ or $\omega\colon \prod_{o:O} \mathbb{R}$, in the type-theoretic sense.

INSIGHT. Sensors that are physically near each other are more likely to see the same color. From this, we can give O a structure by determining the amount of correlation between two sensors, o_α and o_β.

- This justifies assertion that ω is a continuous function; the topology on $\mathbb{R}$ induces a topology on O, and each successive image the camera sees *refines* the topology on O.

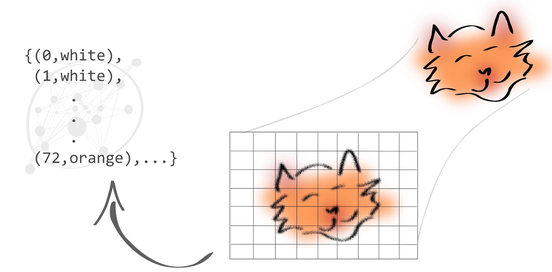

Figure 2. We can think of each photosensor as an *observable*, a type of information we can gain about the outside world. We can think of each image as a *state* of the world. A state acts on an observable, producing a *measured value*.

FEATURES. A structure on O allows for some amount of data compression; if we know that o_α and o_β are near each other, then we don't need both color values to reconstruct essentially *equivalent* images.

We can also *construct features*, for example,

$$o_\alpha \neq o_\beta$$

can be thought of as an edge between o_α and o_β. In fact, one could build a collection O_1 of "higher-order" sensor that detects edges.[2] Continuing inductively, we might end up with an "object sensor."

Abstract Model

SETUP. We have a processor, an outside world, and a family of primitive *observables*, O, bringing in outside information.

- Each observable $o\colon O$ is associated to a type of measured value, which we denote by $V(o)$.
- The outside world presents to us possible *states*, $\omega\colon \Omega$. Each state and observable yields a measurement,

$$\omega(o)\colon V(o).$$

As in the toy model, we think of a state as equivalent to a dependent function, $\omega\colon \prod_{o:O} V(o)$.

We should actually distinguish between "reality", or Ω, and the *state space* S, the collection of all possible combinations of measurements. Two states ω_1 and ω_2 are equivalent if $\omega_1(o) = \omega_2(o)$ for all $o\colon O$ (function extensionality is used to produce an equivalence relation).

SEPARATION. The observables induce a separation (or perhaps, topological distinguishability) between states.

- We can think of the O having the discrete topology.
- The form of equivalence that arises from this is often too strong to be useful or efficient.

WEAKENING. We can weaken the separation if we link together certain observable-value pairs. These links or paths are "abstractions" of certain features in the data. This is analogous to *clustering* used in unsupervised learning, and also perhaps to the construction of *higher inductive types* in type theory.[3]

References

[1] Mitchell, Tom Michael. *The discipline of machine learning*. Vol. 9. Carnegie Mellon University, School of Computer Science, 2006.
[2] Sondhi, Parikshit. "Feature construction methods: a survey." *sifaka.cs.uiuc.edu* 69 (2009): 70-71.
[3] The Univalent Foundations Program. *Homotopy Type Theory: Univalent Foundations of Mathematics*. Institute for Advanced Study, 2013.